SC2006 Software Engineering

Software Requirements Specification

CleanCravings: clean, nearby hawker food that suits your health

Version	2.0 (Lab 5), 2026-11-13
Lab group	SCSL, Group 4
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Revision history

Version	Date	By	Changes
1.0	2026-09	Team	Functional and non-functional requirements (Labs 1 and 2).
1.0.0	2026-09-07	Jovan Chua and Yang Junren	Use case specification finalised.
1.0.1	2026-09-16	Jovan Chua and Yang Junren	Use cases updated after the use case diagram review.
1.1	2026-10-26	Team	UC-0.1 rewritten for Google sign-in.
2.0	2026-11-13	Rafael	Requirements and use case model synchronised with the working prototype for Lab 5: status for every requirement, reworded requirements where no open dataset exists, new FR-0.3, FR-1.8, FR-1.9, FR-5.10, NFR-13 and UC-12.0, Guest and Signed-in Diner actors, and planned use cases marked.

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1. Introduction

1.1 Purpose

This document specifies what CleanCravings does, for the team, the course markers and anyone extending the system. Version 2.0 brings the Lab 1 to 3 requirements and use case model in line with the working prototype demonstrated in Lab 5: each requirement now states whether it is implemented, and wording has been changed where the original assumed data that no open dataset provides.

1.2 Scope

CleanCravings is a responsive web app that helps diners in Singapore choose where to eat at hawker centres. It brings together official data: SFA's SAFE food hygiene grades, NEA's hawker centre list and closure dates, and HPB's Healthier Dining Programme. Anyone can find centres and stalls, filter by hygiene, distance and region, get clean, nearby stall picks and see common hawker dishes rated against HPB-based nutrient thresholds. Signing in with Google adds a health profile, bookmarks, closure alerts and a private visit log. The app gives general guidance; it is not a medical or food safety guarantee.

1.3 Definitions

Term	Meaning
SAFE grade	SFA's Safety Assurance for Food Establishments grade: A, B, C, or NEW for licensees with under a year of record. NEW is not a failing grade.
Track record	A stall's grade history (from the team's dated snapshots of SFA data) and whether its licence was suspended in the past year.
Closure Period	A dated closure of a whole hawker centre from NEA: spring cleaning, repairs and redecoration, or other works.
HDP	HPB's Healthier Dining Programme; participating stalls are "Healthier Dining" stalls.
Dish type	A common hawker dish (41 of them) with typical nutrient values, rated by the rule engine.
Classification rule	One row of the team's rule table: condition, nutrient, operator, threshold, verdict and citation.
Verdict	A dish rating shown as Good choice, Have in moderation, Have occasionally, or Not enough data.
Guest / Signed-in Diner	A user who hasn't signed in / a user signed in with Google.

1.4 References

- Lab 1 and 2: requirements.pdf, Data Dictionary_Updated.pdf, SC2006 UI PROTOTYPE.pdf (course repository).
- Lab 3: SC2006_UseCaseDescription_s4.pdf (use cases v1.0.1), class diagrams and dialog map.
- Lab 4: Lab 4 Test Cases.pdf. Lab 5: class diagrams and dialog map (this submission).
- Data: NEA hawker centres and closure dates, and HPB Healthier Dining partners (data.gov.sg); SFA Track Records portal export.

2. Overall description

2.1 Product perspective

CleanCravings is a new, self-contained system with three parts:

- **Web app** (React and TypeScript): every page, for phones and desktops.
- **Backend API** (Python FastAPI): the only thing the web app talks to (NFR-5). Thin routers pass requests to control classes (hygiene, closures, recommendations, healthier eating, profile, bookmarks, visits, alerts), which read entity classes stored in PostgreSQL on Supabase.
- **Data sync**: one module per source, run daily outside the request path (UC-12.0). A failed source keeps its last good data.

Sign-in is handled by Supabase Auth with Google; the backend only verifies the session token. Maps and place search use OneMap.

2.2 Users

User class	Needs
Guest	Find clean, open centres and stalls nearby; check a stall's grade; see which dishes suit general guidance. No account needed.
Signed-in Diner	Everything a guest can do, plus picks and dish ratings tailored to their health conditions and preferences, bookmarks, closure alerts and a private visit log.

2.3 Operating environment

Current Chrome, Edge, Safari and Firefox on Android, iOS and desktop. The backend runs on Python 3 with PostgreSQL (SQLite for local development and tests).

2.4 Design and implementation constraints

- Only official open data may be shown as fact. SFA has no API, so its data comes from manual exports; history exists only from the first saved snapshot.
- No open dataset gives a stall's dishes, cuisine, prices, contact number, operating hours or demerit points. Requirements that need them are reworded, deferred or planned (see section 3).
- Health guidance must stay general and soft in wording (NFR-6); the app is not a medical device.
- Personal data is handled per Singapore's PDPA (NFR-7, NFR-8).

2.5 Assumptions and dependencies

- data.gov.sg, SFA's portal, OneMap, Google and Supabase remain available and keep their formats.
- SFA licence numbers stay stable across exports, so grades can be linked over time.
- Typical hawker recipes roughly match the published nutrient profiles used for dish ratings.

3. Functional requirements

35 functional requirements: 32 implemented, 1 deferred, 1 withdrawn, 1 partial. "Deferred" means the behaviour is specified but waits for a data source; "Withdrawn" means it no longer applies. IDs from v1 are kept so earlier documents still trace.

Account and session

ID	Requirement (v2.0)	Status	Change from v1	Use cases
FR-0.1	Browsing and searching hawker centres, stalls and dishes, viewing hygiene grades, stall picks and dish ratings shall not require the user to sign in.	**Implemented**	Clarified: lists what guests can do.	UC-2.0, 3.0, 5.0, 6.0
FR-0.2	The system shall require the user to sign in before creating or editing a User Profile, saving a Favourite, logging a Visit, or receiving Closure Alerts.	**Implemented**	Added logging a Visit.	UC-1.0, 4.0, 7.0, 8.0
FR-0.3	The system shall sign users in with Google only, through Supabase Auth, and shall never handle or store passwords.	**Implemented**	New: records the team's sign-in decision.	UC-0.1

Hawker centre and stall discovery

ID	Requirement (v2.0)	Status	Change from v1	Use cases
FR-1.1	The system shall display a list of hawker centres, sourced from NEA's hawker centre dataset, showing name, address, region, number of stalls, today's status and how many stalls hold each SAFE grade.	**Implemented**	Added status and grade counts.	UC-2.0
FR-1.2	The system shall display hawker centres on a map view using each centre's stored geographic coordinates.	**Implemented**	Unchanged.	UC-2.0
FR-1.3	The system shall allow the user to search hawker centres by name, address or postal code, and stalls and dishes by name.	**Implemented**	Cuisine search removed: no open dataset gives a stall's cuisine.	UC-2.0
FR-1.4	The system shall allow the user to filter hawker centres by region.	**Implemented**	Unchanged.	UC-2.0
FR-1.5	The system shall display a hawker centre's cooked-food stalls with each stall's unit number, current SAFE grade and Healthier Dining status. Cuisine, price range and contact number shall be shown when a data source provides them.	**Implemented**	Reworded: no open dataset provides cuisine, price or contact, so the stall page says "Not available yet".	UC-2.0, 3.0
FR-1.6	The system shall display only stalls licensed under the Cooked Food stall type; Market Produce and Piece & Sundries stalls shall not be surfaced.	**Implemented**	Unchanged.	UC-2.0
FR-1.7	The system shall display a stall's operating hours when a data source provides them.	**Deferred**	Reworded: no open dataset publishes stall operating hours.	UC-3.0
FR-1.8	The system shall let the user set their location from the device or by searching a place name or postal code, sort centres by distance and limit results to a chosen distance.	**Implemented**	New: location and distance filter.	UC-2.0, 5.0
FR-1.9	The system shall show each hawker centre's official NEA photo where one is published, served through the team's backend.	**Implemented**	New.	UC-2.0

Food safety and hygiene

ID	Requirement (v2.0)	Status	Change from v1	Use cases
FR-2.1	The system shall display a stall's current SAFE Grade (A, B, C or NEW), sourced from SFA's Safety Assurance for Food Establishments framework.	**Implemented**	Unchanged.	UC-3.0
FR-2.2	The system shall visually distinguish the NEW grade from grade C, consistent with SFA's requirement that NEW not be presented as a failing grade.	**Implemented**	Unchanged.	UC-3.0
FR-2.3	The system shall display a stall's Track Record for the past 12 months: its grade history from dated SFA snapshots and whether its licence was suspended. Demerit points shall be shown as "Not published" until SFA publishes them.	**Implemented**	Reworded: SFA's open data has no demerit points or suspension counts. History starts from the first saved snapshot (20 Oct 2026).	UC-3.0
FR-2.4	The system shall display whether a stall's licence is suspended or was suspended in the past year.	**Implemented**	Clarified to match SFA's "suspended in the past 1 year" flag.	UC-3.0
FR-2.5	The system shall allow the user to filter hawker centres and stalls by a minimum SAFE Grade, ranked A > B > NEW > C. A stall with no current grade shall not pass any minimum.	**Implemented**	Added the ranking and the rule for ungraded stalls.	UC-2.0, 5.0

Closures

ID	Requirement (v2.0)	Status	Change from v1	Use cases
FR-3.1	The system shall display a hawker centre's scheduled Closure Periods with start date, end date and reason (Spring Cleaning, Repairs & Redecoration, or Other Works). Dates NEA marks "TBC" shall not be shown until confirmed.	**Implemented**	Stall-level closures removed: NEA publishes closures per centre only.	UC-2.0, 8.0
FR-3.2	The system shall mark a hawker centre, and the stalls in it, as closed when the current date falls within an active Closure Period, and as closing soon when one starts within 7 days.	**Implemented**	Added "closing soon".	UC-2.0, 3.0
FR-3.3	The system shall exclude stalls at centres that are closed today from Recommendations.	**Implemented**	Unchanged in meaning.	UC-5.0
FR-3.4	The system shall create a Closure Alert for a signed-in user when a hawker centre in their bookmarks has a Closure Period that is active or starts within 7 days, shown in the app's Alerts page. Push notifications are planned.	**Implemented**	Centres only (closures are per centre); alerts are in-app, not push.	UC-8.0

Healthier eating

ID	Requirement (v2.0)	Status	Change from v1	Use cases
FR-4.1	The system shall display a stall's participation in HPB's Healthier Dining Programme and its participation categories, where applicable.	**Implemented**	Unchanged.	UC-3.0
FR-4.2	The system shall rate common hawker dish types against nutrient thresholds based on HPB guidance, for the user's health conditions, showing each rule's threshold and basis. Where a needed nutrient value is missing, it shall say "Not enough data" instead of guessing.	**Implemented**	Replaces "Lower Calorie Dish status": no dataset links hawker dishes to that label.	UC-6.0
FR-4.3	Withdrawn: display an ingredient's Healthier Choice Symbol.	**Withdrawn**	The symbol applies to packaged products; no dataset links it to hawker food.	-
FR-4.4	The system shall not imply that a stall outside the Healthier Dining Programme serves unhealthy food, consistent with the programme's voluntary nature.	**Implemented**	Unchanged.	UC-3.0

User profile and personalisation

ID	Requirement (v2.0)	Status	Change from v1	Use cases
FR-5.1	The system shall allow a signed-in user to set Dietary Restrictions. No pork, no beef and vegetarian shall exclude dish types that don't match. Halal shall show only stalls whose halal status HPB has confirmed; an unknown status shall never be treated as non-halal.	**Implemented**	Reworded: no dataset says which dishes a stall sells, so only halal can filter stalls.	UC-1.0, 2.0, 5.0, 6.0
FR-5.2	The system shall allow a signed-in user to set Dietary Preferences (prefer Healthier Dining stalls; lower sodium first) that change ranking without excluding results.	**Implemented**	Preferred cuisine removed (no cuisine data).	UC-1.0, 5.0, 6.0
FR-5.3	The system shall allow a signed-in user to set a Budget Range (minimum and maximum price per meal).	**Partial**	Saved, but not used until a source provides stall prices.	UC-1.0
FR-5.4	The system shall allow a signed-in user to set a Hygiene Threshold (minimum acceptable SAFE Grade).	**Implemented**	Unchanged.	UC-1.0
FR-5.5	The system shall allow a signed-in user to set one or more preferred regions.	**Implemented**	Unchanged.	UC-1.0
FR-5.6	The system shall generate a ranked list of stall picks for any user from open centres, by SAFE grade and distance. For a signed-in user it shall also apply their hygiene threshold and halal restriction, and favour their preferred regions, Healthier Dining preference, bookmarks and well-rated visits. Each pick shall show why it is listed.	**Implemented**	Guests now get picks too; ranking inputs listed.	UC-5.0
FR-5.7	The system shall allow a signed-in user to save a hawker centre, stall or dish as a Favourite, view them together and remove them.	**Implemented**	Dishes added.	UC-7.0
FR-5.8	The system shall allow a signed-in user to record a Visit to a stall, with an optional rating, dish and note.	**Implemented**	Added the optional dish.	UC-4.0
FR-5.9	The system shall use a user's visit ratings in future Recommendations (rated 4 or 5: higher; 1 or 2: lower).	**Implemented**	Was a stretch goal.	UC-4.0, 5.0
FR-5.10	The system shall allow a signed-in user to record health conditions (high blood pressure, diabetes, high cholesterol), used only to rate dishes for them.	**Implemented**	New: health conditions were in the use cases but had no requirement.	UC-1.0, 6.0

4. Non-functional requirements

13 non-functional requirements. NFR-13 is new in v2.0.

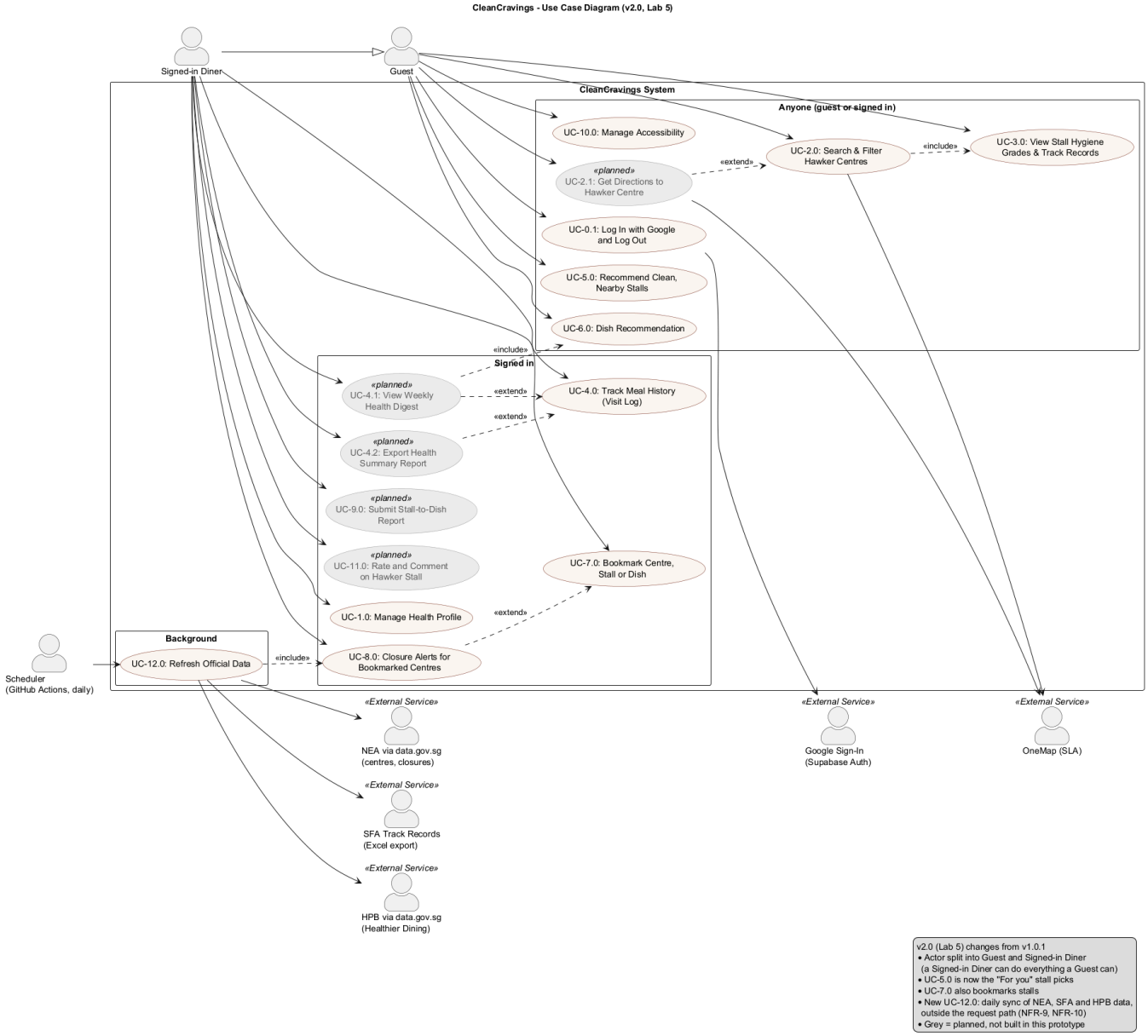
ID	Category	Requirement	Status	How the prototype meets it
NFR-1	Performance	The hawker centre list and map view shall load within 3 seconds on standard mobile data, for about 123 hawker centres.	Partial	The centre list and the data behind picks are built once per data version and cached in memory, so after start-up a request needs one small query. Load time on mobile data hasn't been formally measured.
NFR-2	Data freshness	The system shall show a "last updated" time for SAFE Grades, Track Records and Healthier Dining status.	Implemented	Per-source last-updated times (GET /meta/freshness) and a source line on each stall page.
NFR-3	Usability	SAFE Grades, Track Record summaries, closure status and Healthier Eating tags shall be visually distinguishable at a glance.	Implemented	Separate badge styles for grades, status pills, Healthier Dining tags and verdicts.
NFR-4	Data provenance	Hygiene data shall be sourced from SFA, closures from NEA, and Healthier Dining data from HPB, and cited as such.	Implemented	Each page cites its source; the footer lists all of them.
NFR-5	System architecture	All hygiene, closure and healthier-eating data shall reach the client through the team's own backend API, not directly from files or third-party endpoints.	Implemented	The frontend has one HTTP client (lib/api.ts). Geocoding and NEA photos are also proxied.
NFR-6	Accuracy and honesty	The system shall not overstate the precision of a recommendation or present it as a guarantee of food safety or suitability.	Implemented	The score is never shown; verdicts use soft wording; picks and ratings carry a disclaimer.
NFR-7	Privacy	Health conditions, Dietary Restrictions, Dietary Preferences and Visit history shall be stored only as needed for personalisation and never shared with third parties.	Implemented	Stored only in the team's database; row-level security blocks Supabase's public API from every table.
NFR-8	Security	User accounts and profile data shall use encrypted storage and secure credential handling, per Singapore's PDPA.	Partial	Google sign-in, so no passwords are stored; the backend verifies every token. Supabase encrypts data at rest. HTTPS is needed when deployed (the prototype runs locally).
NFR-9	Reliability	On a source dataset outage, the system shall degrade gracefully using cached data with a staleness indicator.	Implemented	A failed sync rolls back and keeps the last good data; data older than 48 hours shows an "Offline mode" banner.
NFR-10	Maintainability	Each data source (NEA, SFA, HPB) shall be integrated as an independent, updatable module.	Implemented	One sync module per source, each runnable on its own and wrapped in the same sync_run logging.
NFR-11	Compatibility	The system shall be usable on both Android and iOS mobile browsers.	Implemented	One responsive web app with a bottom tab bar on phones; checked at phone screen sizes.
NFR-12	Scalability	The architecture shall scale to all hawker centres and their stalls without redesign.	Implemented	Runs on all 123 centres and about 5,600 cooked-food stalls.

NFR-13	Accessibility	The system shall offer dark mode and at least four text sizes, available without signing in and applied without reloading.	**Implemented**	New, as UC-10.0 recommended. Settings are saved on the device.

5. Use case model

5.1 Use case diagram

The actor is split into Guest and Signed-in Diner, who can do everything a Guest can. Grey use cases are planned and not built in this prototype. The data sources are now actors of UC-12.0, the daily sync, rather than of the use cases that display their data.



Also submitted as a separate image: CleanCravings_UseCaseDiagram.jpg.

5.2 Use case summary

ID	Use case	Primary actor	Status
UC-0.1	Log In with Google and Log Out	Guest	Implemented
UC-1.0	Manage Health Profile	Signed-in Diner	Implemented
UC-2.0	Search and Filter Hawker Centres	Guest	Implemented
UC-2.1	Get Directions to Hawker Centre	Guest; OneMap routing	Planned
UC-3.0	View Stall Hygiene Grades and Track Records	Guest	Implemented
UC-4.0	Track Meal History (Visit Log)	Signed-in Diner	Implemented
UC-4.1	View Weekly Health Digest	Signed-in Diner	Planned
UC-4.2	Export Health Summary Report	Signed-in Diner	Planned
UC-5.0	Recommend Clean, Nearby Stalls	Guest	Implemented
UC-6.0	Dish Recommendation	Guest	Implemented
UC-7.0	Bookmark Hawker Centre, Stall or Dish	Signed-in Diner	Implemented
UC-8.0	Closure Alerts for Bookmarked Centres	Signed-in Diner	Implemented
UC-9.0	Submit Stall-to-Dish Report	Signed-in Diner	Planned
UC-10.0	Manage Accessibility	Guest	Implemented
UC-11.0	Rate and Comment on Hawker Stall	Signed-in Diner	Planned
UC-12.0	Refresh Official Data	Scheduler (GitHub Actions, daily at 02:00 Singapore time)	Implemented

5.3 Use case descriptions (built)

Template as in v1 (Karl E. Wiegers). Each description states what changed from v1.0.1.

UC-0.1 Log In with Google and Log Out

Use Case ID	UC-0.1
Use Case Name	Log In with Google and Log Out
Status	**Implemented**
Created By	Jovan Chua, 2026-09-04
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Guest. Secondary: Google Sign-In (via Supabase Auth).
Description	Allows a user to sign in with their Google account so they can save a health profile, bookmarks and visits, and receive closure alerts, and to sign out again. CleanCravings never sees or stores a password.
Preconditions	1. The user has a Google account. 2. The device has a network connection.
Postconditions	1. The user is signed in, and the browser holds a session that is sent to the backend as a Bearer token. 2. On first sign-in, the system creates the user's account record.
Priority	High
Frequency of Use	High
Flow of Events	1. User selects "Sign in" in the header, or an action that needs an account (for example Save or Log a visit). 2. System displays the Sign-in screen. 3. User selects "Continue with Google". 4. System redirects the user to Google through Supabase Auth. 5. User chooses a Google account and approves access. 6. Supabase Auth returns the user to CleanCravings with a session, which the browser stores. 7. Backend verifies the session token and creates the user's account record on first sign-in. 8. If the user has no health profile yet, System opens profile setup (UC-1.0); otherwise it opens Home.
Alternative Flows	**0.1.AC.1 Returning session:** At Step 1, if a valid session is stored, the system signs the user in automatically. **0.1.AC.2 Continue as guest:** At Step 2, the user selects "Continue as guest" and keeps browsing without an account (FR-0.1). **0.1.AC.3 Sign out:** From Profile, the user selects "Sign out". The system ends the session and removes it from the device.
Exceptions	**0.1.EX.1 Sign-in cancelled:** At Step 5, if the user cancels, no session is created and they stay signed out. **0.1.EX.2 Session rejected:** If the backend can't verify the token, the system treats the user as signed out. **0.1.EX.3 Sign-in service unavailable:** At Step 4, if Google or Supabase can't be reached, sign-in fails and the user can keep browsing as a guest.
Includes / Extends	None
Special Requirements	No passwords are handled or stored (NFR-8). Signing out removes the stored session, so a shared device doesn't stay signed in. Traces to FR-0.1, FR-0.2, FR-0.3, NFR-8.
Assumptions	The Google provider is enabled in the team's Supabase project.

Notes and Changes from v1	v1 described email and password sign-in with "Remember me" and "Forgot password". These were dropped when the team chose Google sign-in.

UC-1.0 Manage Health Profile

Use Case ID	UC-1.0
Use Case Name	Manage Health Profile
Status	**Implemented**
Created By	Jovan Chua, 2026-09-04
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Signed-in Diner.
Description	Allows a signed-in user to record their health conditions, dietary restrictions and preferences, minimum SAFE grade, preferred regions and budget, so that stall picks and dish ratings are tailored to them.
Preconditions	1. The user is signed in (UC-0.1).
Postconditions	1. The profile is saved. 2. Stall picks (UC-5.0), dish ratings (UC-6.0) and centre stall lists use the new profile from the next request.
Priority	Medium
Frequency of Use	Low (at first sign-in and when needs change)
Flow of Events	1. User opens profile setup after first sign-in, or selects Edit on the Profile page. 2. System loads any saved profile and shows the form: health conditions, dietary restrictions, preferences, minimum SAFE grade, preferred regions and budget. 3. User selects the options that apply. 4. User selects "Save profile". 5. System validates and saves the profile. 6. System returns to Home (first time) or Profile, where the picks and ratings now follow the profile.
Alternative Flows	**1.0.AC.1 Update an existing profile:** At Step 2, the form is pre-filled with the saved values; the user changes some and continues at Step 4. **1.0.AC.2 Skip:** During first setup, the user selects "Skip". No profile is saved and general guidance is used. **1.0.AC.3 Leave with unsaved changes:** Before Step 4, the user selects Cancel. If the form changed, the system asks "Discard unsaved changes?" first.
Exceptions	**1.0.EX.1 Save fails:** At Step 5, if the server can't be reached or returns an error, the system keeps the form as it is and shows the error. **1.0.EX.2 Invalid value:** At Step 5, if a value isn't allowed (for example an unknown region), the system rejects the save and shows the error.
Includes / Extends	None
Special Requirements	Health conditions are used only to rate dishes and are never shared (NFR-7). The form says the ratings are general guidance, not medical advice. Traces to FR-5.1 to FR-5.5, FR-5.10, NFR-7.
Assumptions	Users choose conditions that apply to them; the app doesn't verify them.
Notes and Changes from v1	v1's "Healthier Choice" option is now the "Prefer Healthier Dining stalls" preference. The budget is saved but not used until a source provides stall prices. Offline caching of the form was not built.

UC-2.0 Search and Filter Hawker Centres

Use Case ID	UC-2.0
Use Case Name	Search and Filter Hawker Centres
Status	**Implemented**

Created By	Jovan Chua, 2026-09-04
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Guest. Secondary: OneMap (map tiles and place search, through the backend).
Description	Allows a user to find hawker centres by name, place or distance, filter them by region and minimum SAFE grade, see them on a map, and open a centre to see its cooked-food stalls.
Preconditions	1. Official data has been loaded at least once (UC-12.0).
Postconditions	1. The matching centres are shown as a list or on the map; a selected centre's page shows its stalls.
Priority	High
Frequency of Use	High
Flow of Events	1. User opens Finder & Map. 2. System lists all hawker centres with photo, today's status, SAFE grade counts and Healthier Dining count. 3. User selects "Use my location", or searches a place name or postal code and picks a result. 4. System sorts the centres by distance and shows each distance. 5. User filters by region and minimum SAFE grade, and drags the distance slider. The list updates straight away; the distance filter needs no new request. 6. User switches to Map. System shows the same centres on OneMap, coloured by status. 7. User types in the search box. System narrows the centres by name, address or postal code, and lists matching stalls and dishes. 8. User selects a centre. System shows its photo, status, address, upcoming closures, a map, directions links, and its cooked-food stalls with SAFE grade and Healthier Dining status. 9. User narrows the stalls by minimum grade, Healthier Dining only, or confirmed halal only. 10. User selects a stall (UC-3.0).
Alternative Flows	**2.0.AC.1 Location permission denied:** At Step 3, the user searches a place or postal code instead. **2.0.AC.2 No location set:** Steps 3 and 4 are skipped; the list isn't sorted by distance and the slider is hidden. **2.0.AC.3 Halal restriction in profile:** At Step 8, for a signed-in user with the halal restriction, the stall list starts with "Confirmed halal only" on, and says that most stalls haven't published a status.
Exceptions	**2.0.EX.1 Location not found:** At Step 3, if the place search finds nothing, the system shows "Location not found. Try again." **2.0.EX.2 No centres match:** At Step 5 or 7, the system shows "No centres match. Try removing a filter." **2.0.EX.3 No stalls match:** At Step 9, the system shows "No matches found." with how many stalls the filters hid. **2.0.EX.4 Stale data:** If a source hasn't synced for 48 hours, a banner says the app is showing previous data (NFR-9).
Includes / Extends	UC-3.0 View Stall Hygiene Grades and Track Records. Extended by UC-2.1 (planned).
Special Requirements	All data comes through the team's backend (NFR-5). Traces to FR-1.1 to FR-1.6, FR-1.8, FR-1.9, FR-2.5, FR-3.1, FR-3.2, FR-5.1, NFR-1.
Assumptions	Users are in Singapore.
Notes and Changes from v1	v1 described an offline SQLite snapshot on the device; instead the server keeps the last good data. Filtering stalls by health condition moved to dish ratings (UC-6.0), because no dataset says which dishes a stall sells.

UC-3.0 View Stall Hygiene Grades and Track Records

Use Case ID	UC-3.0
Use Case Name	View Stall Hygiene Grades and Track Records

Status	**Implemented**
Created By	Yang Junren, 2026-09-07
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Guest. (SFA and HPB data arrive through UC-12.0.)
Description	Allows a user to see a stall's official SAFE grade explained in plain words, its track record, Healthier Dining status and where the data comes from.
Preconditions	1. SFA data has been imported at least once (UC-12.0).
Postconditions	1. The stall page shows the grade, track record and sources.
Priority	High
Frequency of Use	High
Flow of Events	1. User selects a stall from a centre page, search results, stall picks or Saved. 2. System shows the stall name, centre and unit, and a banner if the whole centre is closed or closing soon. 3. System shows the SAFE grade badge with its meaning (NEW: open under a year, not a failing grade), any "under review" flag, and the source with its last-updated date. 4. System shows Healthier Dining status and endorsement criteria, or "Not listed" with a note that the programme is voluntary; halal status appears only if HPB published it. 5. System shows the track record: licence suspended in the past 12 months, demerit points ("Not published"), licensee, and grade history from dated snapshots. 6. System shows stall info that no dataset provides yet (cuisine, price, contact, hours) as "Not available yet". 7. A signed-in user can save the stall (UC-7.0) or log a visit (UC-4.0).
Alternative Flows	**3.0.AC.1 No grade on record:** At Step 3, the system shows "No grade on record from SFA". Such stalls don't pass any minimum-grade filter.
Exceptions	**3.0.EX.1 Stall not found:** At Step 2, if the stall doesn't exist, the system shows an error with a retry option.
Includes / Extends	None
Special Requirements	NEW must not look like a failing grade (FR-2.2). A SAFE grade is SFA's assessment of track record, not a guarantee. Traces to FR-1.5, FR-1.7, FR-2.1 to FR-2.4, FR-4.1, FR-4.4, NFR-2 to NFR-4.
Assumptions	SFA licence numbers link grades to stalls across exports.
Notes and Changes from v1	v1 listed demerit points with violation reasons and per-stall dietary flags. SFA's open data has neither, so the page says "Not published", and dietary suitability is shown per dish (UC-6.0).

UC-4.0 Track Meal History (Visit Log)

Use Case ID	UC-4.0
Use Case Name	Track Meal History (Visit Log)
Status	**Implemented**
Created By	Yang Junren, 2026-09-07
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Signed-in Diner.
Description	Allows a signed-in user to log a visit to a stall, with an optional rating, dish and note, and to see their visit history. Ratings feed into future stall picks.
Preconditions	1. The user is signed in. 2. The user is on a stall page (UC-3.0).
Postconditions	1. The visit is stored with today's date. 2. Future picks (UC-5.0) favour stalls rated 4 or 5 and lower those rated 1 or 2.

Priority	Low
Frequency of Use	Medium (about once a day)
Flow of Events	1. User selects "Log a visit" on a stall page. 2. System shows the form: a 1 to 5 star rating, the dish eaten (from the dish list) and a note of up to 300 characters, all optional. 3. User fills in what they want and selects "Save visit". 4. System stores the visit and shows "Visit logged. It'll nudge your future picks." 5. User opens Profile. System lists the visit history, newest first.
Alternative Flows	**4.0.AC.1 Guest:** At Step 1, a user who isn't signed in is taken to the Sign-in screen (UC-0.1). **4.0.AC.2 Cancel:** At Step 3, the user selects Cancel and nothing is saved.
Exceptions	**4.0.EX.1 Invalid rating:** A rating outside 1 to 5 is rejected. **4.0.EX.2 Save fails:** At Step 4, if the server can't be reached, the system shows the error and keeps the form.
Includes / Extends	None (the ratings are read when picks are generated, UC-5.0). Extended by UC-4.1 and UC-4.2 (planned).
Special Requirements	Visits are private to the user (NFR-7). Traces to FR-0.2, FR-5.8, FR-5.9.
Assumptions	Users log visits honestly; the app doesn't verify them.
Notes and Changes from v1	v1's meal slot (breakfast, lunch and so on), quick-log and offline queue were not built.

UC-5.0 Recommend Clean, Nearby Stalls

Use Case ID	UC-5.0
Use Case Name	Recommend Clean, Nearby Stalls
Status	**Implemented**
Created By	Jovan Chua, 2026-09-08
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Guest.
Description	Shows the "For you" list: stalls from centres that are open today, ranked by SAFE grade and distance, and for signed-in users also by their profile, bookmarks and visit ratings. Each pick says why it is listed.
Preconditions	1. Official data has been loaded (UC-12.0).
Postconditions	1. Up to 20 picks are shown, at most 3 from any one centre, with reasons and a note on what was left out.
Priority	High
Frequency of Use	High (every visit to Home)
Flow of Events	1. User opens For you (Home). 2. User optionally sets a location (as in UC-2.0). 3. System takes the cooked-food stalls of every centre, and leaves out centres that are closed today or not operating, and centres more than 8 km away if a location is set. 4. For a signed-in user with a profile, System also leaves out stalls below their minimum SAFE grade and, with the halal restriction, stalls without a confirmed halal status. 5. System ranks the rest by distance, SAFE grade, Healthier Dining (more if preferred), preferred region, bookmarks and the user's own visit ratings. 6. System shows the top picks, each with plain reasons such as "3.3 km away", "In your preferred region (West)", "You rated it 5/5" or "Heads up: centre closes in 5 day(s)". 7. System notes how many stalls were left out because their centre is closed today, by the hygiene threshold or by the halal filter, and that picks are suggestions, not guarantees.

Alternative Flows	**5.0.AC.1 Guest or no profile:** Step 4 is skipped. Home invites signed-in users without a profile to set one up. **5.0.AC.2 No location:** Distance isn't used; picks are ranked on the other factors.
Exceptions	**5.0.EX.1 No matches:** The system shows "No matches found." and suggests another location or a lower hygiene threshold.
Includes / Extends	None.
Special Requirements	The score is never shown (NFR-6). Centre data is cached per data version so picks stay fast (NFR-1). Traces to FR-0.1, FR-3.3, FR-5.1, FR-5.2, FR-5.4 to FR-5.6, FR-5.9.
Assumptions	Distance is straight-line (haversine), not travel distance.
Notes and Changes from v1	v1 was "Update Recommendation Engine", a background update of a local cache weighted by meal frequency. The picks are now computed on the server for each request, from the inputs above, so it is renamed and shown as a use case the user starts.

UC-6.0 Dish Recommendation

Use Case ID	UC-6.0
Use Case Name	Dish Recommendation
Status	**Implemented**
Created By	Jovan Chua, 2026-09-08
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Guest.
Description	Rates common hawker dish types against nutrient thresholds based on HPB guidance, for the user's health conditions, and explains each rating.
Preconditions	1. Dish profiles and rules have been loaded (UC-12.0).
Postconditions	1. Dish types are shown grouped by verdict, each with the rules behind it.
Priority	High
Frequency of Use	High
Flow of Events	1. User opens Dishes. 2. System uses the health conditions and restrictions from the user's profile if signed in, otherwise none. 3. System evaluates each dish type against the general rules and the rules for each condition. The worst fired rule decides the verdict. 4. System groups the dishes as Good choice, Have in moderation, Not enough data and Have occasionally, with counts. 5. User opens a dish. System shows its nutrients, each fired rule with its threshold, rule ID and basis, any data gaps, the data source and the photo credit. 6. User switches conditions or restrictions on or off to see other ratings; this doesn't change the saved profile. 7. User opens "How we rate dishes" to see every rule.
Alternative Flows	**6.0.AC.1 No conditions:** At Step 2, only the general rules apply, and the page invites the user to add conditions. **6.0.AC.2 Lower sodium first:** With this preference, dishes within each group are ordered by sodium. **6.0.AC.3 Bookmark a dish:** A signed-in user saves a dish (UC-7.0).
Exceptions	**6.0.EX.1 Nothing matches:** If the restrictions leave no dishes, the system shows "No dishes match all selected constraints." with an "Expand search" button that clears them. **6.0.EX.2 Missing nutrient:** If a value needed for one of the user's conditions is missing, the verdict is "Not enough data", unless another rule already says Have occasionally.
Includes / Extends	None (uses the profile from UC-1.0 when there is one).

Special Requirements	Wording is soft ("Have occasionally", never "unhealthy") and ratings are for typical recipes, not a particular stall (NFR-6). Rules are data: changing a threshold means editing a CSV row, not code. Traces to FR-4.2, FR-5.1, FR-5.2, FR-5.10, NFR-6.
Assumptions	Typical hawker recipes roughly match the published nutrient profiles.
Notes and Changes from v1	v1's "Find this nearby" link wasn't built, because no dataset says which stalls sell a dish (see UC-9.0). The 13 rule thresholds are marked draft until the team confirms each against a published HPB guideline; some citations still say TBC.

UC-7.0 Bookmark Hawker Centre, Stall or Dish

Use Case ID	UC-7.0
Use Case Name	Bookmark Hawker Centre, Stall or Dish
Status	**Implemented**
Created By	Yang Junren, 2026-09-16
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Signed-in Diner.
Description	Allows a signed-in user to save hawker centres, stalls and dish types, see them in one place, and remove them.
Preconditions	1. The user is signed in. 2. The user is on a centre page, a stall page or the Dishes page.
Postconditions	1. The item is added to, or removed from, the user's bookmarks. 2. A saved centre is checked for closures (UC-8.0).
Priority	Medium
Frequency of Use	Medium (a few times a week)
Flow of Events	1. User selects Save on a centre or stall page, or the bookmark icon on a dish. 2. System saves the item with its type and time, fills the icon, and confirms (for a centre: "Saved. You'll get closure alerts for it."). 3. User opens Saved. System groups the bookmarks into centres, stalls and dishes, with today's status for centres and stalls. 4. User selects a bookmark and System opens its page.
Alternative Flows	**7.0.AC.1 Remove:** The user selects the icon again, or the remove button on Saved, and the bookmark is deleted. **7.0.AC.2 Guest:** At Step 1, a user who isn't signed in is taken to the Sign-in screen.
Exceptions	**7.0.EX.1 No longer listed:** At Step 3, a bookmarked centre or stall that has left the official data is shown as "Unavailable: no longer in the official list". **7.0.EX.2 Save fails:** At Step 2, if the server can't be reached, the system shows the error.
Includes / Extends	None. Extended by UC-8.0.
Special Requirements	Traces to FR-0.2, FR-5.7.
Assumptions	Bookmarks are kept in the user's account, so they appear on any device after sign-in.
Notes and Changes from v1	v1 said bookmarks don't affect recommendations; they now give a small boost to picks (FR-5.6). Stalls can be bookmarked too. Offline saving wasn't built.

UC-8.0 Closure Alerts for Bookmarked Centres

Use Case ID	UC-8.0
Use Case Name	Closure Alerts for Bookmarked Centres
Status	**Implemented**
Created By	Jovan Chua, 2026-09-18

Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Signed-in Diner. Secondary: NEA closure data (through UC-12.0).
Description	Alerts a user in the app when a hawker centre they saved is closed or will close within 7 days for cleaning or works.
Preconditions	1. The user is signed in and has saved at least one centre (UC-7.0). 2. NEA closure dates have been synced (UC-12.0).
Postconditions	1. One alert exists per user per closure. 2. The centre shows as closed or closing soon, and closed centres are left out of picks (FR-3.3).
Priority	Medium
Frequency of Use	Low (when a saved centre has a closure)
Flow of Events	1. The check runs when the user saves a centre, and after each daily data sync. 2. System finds closures of the user's saved centres that are active or start within 7 days. 3. System creates an alert for each new one, for example "Tiong Bahru Market will be closed from 16 Nov to 16 Nov for Spring Cleaning." 4. The bell in the header shows the number of unread alerts. 5. User opens Alerts. System lists each alert with its reason, dates and when it was sent. 6. User selects an alert. System opens the centre page and marks the alert read.
Alternative Flows	**8.0.AC.1 Mark read:** At Step 5, the user selects "Mark read" without opening the centre. **8.0.AC.2 Closure over:** After the end date, the centre shows as open again and its stalls return to picks; the alert stays in the list.
Exceptions	**8.0.EX.1 NEA unavailable:** If the NEA sync fails, the last good closure data is used and no new alerts are made until it succeeds. **8.0.EX.2 Date TBC:** Closures NEA marks "TBC" aren't imported, so they don't produce alerts until confirmed.
Includes / Extends	Extends UC-7.0. Included by UC-12.0 (alerts after each sync).
Special Requirements	Alerts only for centres the user saved (no unsolicited alerts). Traces to FR-3.1 to FR-3.4.
Assumptions	NEA's closure dataset reflects real closures at each sync.
Notes and Changes from v1	Push notifications and "open again" alerts are planned; alerts are in-app for now. NEA publishes closures per centre, not per stall.

UC-10.0 Manage Accessibility

Use Case ID	UC-10.0
Use Case Name	Manage Accessibility
Status	**Implemented**
Created By	Jovan Chua, 2026-09-18
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Guest.
Description	Allows any user to choose light or dark mode and a text size, to make the app easier to read.
Preconditions	1. The app is open.
Postconditions	1. The chosen theme and text size apply across the app straight away and are remembered on the device.
Priority	Medium
Frequency of Use	Low

Flow of Events	1. User opens Profile and goes to Accessibility. 2. System shows the theme (Light, Dark, Match device) and text size (Small, Default, Large, Extra large). 3. User selects a theme. System re-colours the app at once; the map switches to OneMap's night tiles in dark mode. 4. User selects a text size. System resizes all text at once.
Alternative Flows	**10.0.AC.1 Match device:** At Step 3, the app follows the device's light or dark setting and changes when it does. **10.0.AC.2 Guest:** Guests can change these settings too; they are saved on the device.
Exceptions	**10.0.EX.1 Storage blocked:** If the browser blocks storage, the settings apply for this visit only.
Includes / Extends	None
Special Requirements	No reload needed; text stays legible in both themes at every size. Traces to NFR-11, NFR-13.
Assumptions	The browser reports the device's light or dark setting.
Notes and Changes from v1	Voice search and syncing settings to the account were not built.

UC-12.0 Refresh Official Data

Use Case ID	UC-12.0
Use Case Name	Refresh Official Data
Status	**Implemented**
Created By	Rafael, 2026-11-13 (new in v2.0)
Last Updated	Rafael, 2026-11-13 (v2.0)
Actors	Primary: Scheduler (GitHub Actions, daily at 02:00 Singapore time). Secondary: NEA and HPB datasets on data.gov.sg, SFA Track Records export.
Description	Loads the latest official data into the database outside the request path, one source at a time, and creates closure alerts. A failed source keeps its last good data.
Preconditions	1. The database is reachable. 2. For SFA: a new export from SFA's Track Records portal has been placed in etl/inbox (SFA has no API).
Postconditions	1. Each source's run is logged with its status, time and row count, which the app shows as "last updated". 2. New closure alerts have been created (UC-8.0).
Priority	High
Frequency of Use	Daily (and by hand when a new SFA export arrives)
Flow of Events	1. Scheduler starts the daily run. 2. System loads the team's dish profiles and classification rules from CSV. 3. System loads hawker centres and closure dates from NEA on data.gov.sg. 4. If a new SFA export is in the inbox, System imports SAFE grades for stalls at hawker centres as a new dated snapshot. 5. System loads HPB's Healthier Dining partners and links them to stalls by name (exact, then a guarded fuzzy match, then manual overrides, otherwise centre-level), and takes halal status from the listing. 6. System creates closure alerts for saved centres (UC-8.0). 7. System records each source's result; the app's caches refresh on the next request.
Alternative Flows	**12.0.AC.1 No new SFA export:** At Step 4, SFA is skipped and the previous snapshot stays current. **12.0.AC.2 Run one source by hand:** A team member runs a single source module on its own.

Exceptions	**12.0.EX.1 Source fails:** The failed source rolls back and keeps its last good data; the other sources still run. **12.0.EX.2 Data goes stale:** If a source hasn't succeeded for 48 hours, the app shows an "Offline mode" banner (NFR-9).
Includes / Extends	UC-8.0 Closure Alerts (Step 6).
Special Requirements	Each source is an independent module (NFR-10); the app only reads what is already in the database (NFR-5). Traces to NFR-2, NFR-4, NFR-9, NFR-10, FR-3.4.
Assumptions	NEA, SFA and HPB keep their dataset formats and licence numbers stable.
Notes and Changes from v1	New in v2.0, because the data sync is a major part of the system and wasn't in the v1 model.

5.4 Planned use cases

These remain in the model as agreed future work. Their full v1.0.1 descriptions are in *lab3/SC2006_UseCaseDescription_s4.pdf* and still apply; a summary is below.

ID	Use case	Intended behaviour
UC-2.1	**Get Directions to Hawker Centre** By Yang Junren. Actor: Guest; OneMap routing. Extends UC-2.0.	Show a route, travel time and step-by-step directions to a centre for a chosen travel mode. For now the centre page links to Google Maps and OneMap instead (v1 alternative 2.1.AC.3).
UC-4.1	**View Weekly Health Digest** By Yang Junren. Actor: Signed-in Diner. Extends UC-4.0; includes UC-6.0.	Group the past 7 days of logged visits by dish verdict, name the most common concern and suggest dish types for the coming week.
UC-4.2	**Export Health Summary Report** By Yang Junren. Actor: Signed-in Diner. Extends UC-4.0.	Export logged meals for a date range as a PDF, with health conditions left out unless the user includes them.
UC-9.0	**Submit Stall-to-Dish Report** By Yang Junren. Actor: Signed-in Diner. None.	Let users report which dish types a stall sells, shown as community-reported, rate-limited and flaggable. This would enable "Find this nearby" in UC-6.0 and per-stall dish ratings.
UC-11.0	**Rate and Comment on Hawker Stall** By Jovan Chua. Actor: Signed-in Diner. None.	Public star ratings and comments on stalls, labelled as community content and moderated. Today's visit ratings (UC-4.0) are private.

6. Traceability: requirements to use cases and code

File and class names refer to the source code in the submission. Automated tests (pytest) cover the hygiene rules, rule engine, closure status, API endpoints and data parsers; see *Lab 4 Test Cases.pdf*.

Requirement	Status	Use cases	Implemented in
FR-0.1	Implemented	UC-2.0, 3.0, 5.0, 6.0	`routers/public.py`
FR-0.2	Implemented	UC-1.0, 4.0, 7.0, 8.0	`routers/me.py, auth.require_user`
FR-0.3	Implemented	UC-0.1	`app/auth.py, frontend lib/auth.tsx`
FR-1.1	Implemented	UC-2.0	`HawkerCentreController.list_centres, Finder page`
FR-1.2	Implemented	UC-2.0	`MapView (Leaflet with OneMap tiles)`
FR-1.3	Implemented	UC-2.0	`HawkerCentreController.search, Finder page`
FR-1.4	Implemented	UC-2.0	`list_centres(region)`
FR-1.5	Implemented	UC-2.0, 3.0	`centre_detail, CentreDetail page`
FR-1.6	Implemented	UC-2.0	`sfa_import (cooked food only), _cooked()`
FR-1.7	Deferred	UC-3.0	`Stall page shows "Not available yet"`
FR-1.8	Implemented	UC-2.0, 5.0	`/api/geocode (OneMap), LocationBar, Finder radius slider`
FR-1.9	Implemented	UC-2.0	`HawkerCentreController.photo, GET /centres/{id}/photo`
FR-2.1	Implemented	UC-3.0	`HygieneController.current_grade, GradeBadge`
FR-2.2	Implemented	UC-3.0	`GRADE_RANK (A > B > NEW > C), labels.ts`
FR-2.3	Implemented	UC-3.0	`HygieneController.track_record`
FR-2.4	Implemented	UC-3.0	`HygieneController.suspended_past_year`
FR-2.5	Implemented	UC-2.0, 5.0	`HygieneController.grade_passes`
FR-3.1	Implemented	UC-2.0, 8.0	`nea_sync, CentreDetail page`
FR-3.2	Implemented	UC-2.0, 3.0	`ClosureController.centre_status`
FR-3.3	Implemented	UC-5.0	`RecommendationController.recommend`
FR-3.4	Implemented	UC-8.0	`ClosureController.generate_alerts, Alerts page`
FR-4.1	Implemented	UC-3.0	`hpb_partners_sync, HdpBadge`
FR-4.2	Implemented	UC-6.0	`rules/engine.evaluate, classification_rules.csv`
FR-4.3	Withdrawn	-	-
FR-4.4	Implemented	UC-3.0	`HDP_NOTE on centre and stall pages`
FR-5.1	Implemented	UC-1.0, 2.0, 5.0, 6.0	`presenters.confirmed_halal, _restricted()`
FR-5.2	Implemented	UC-1.0, 5.0, 6.0	`recommend(), assess_dishes(prefer_lower_sodium)`
FR-5.3	Partial	UC-1.0	`UserProfile.budget_min / budget_max`
FR-5.4	Implemented	UC-1.0	`UserProfile.hygiene_threshold`
FR-5.5	Implemented	UC-1.0	`UserProfile.preferred_regions`
FR-5.6	Implemented	UC-5.0	`RecommendationController.recommend`
FR-5.7	Implemented	UC-7.0	`FavouriteController, Saved page`
FR-5.8	Implemented	UC-4.0	`VisitController, Log a visit`
FR-5.9	Implemented	UC-4.0, 5.0	`recommend()`
FR-5.10	Implemented	UC-1.0, 6.0	`UserProfile.health_conditions`

7. Open issues

Issue	Affects	Next step
All 13 dish rule thresholds are marked draft; some citations still say TBC.	FR-4.2, UC-6.0	Confirm each threshold against a published HPB guideline, then mark it reviewed in the rules CSV.
No open data on a stall's dishes, cuisine, prices, contact or hours.	FR-1.5, FR-1.7, FR-5.3, UC-9.0	Community reports (UC-9.0) or a future official dataset.
Closure alerts are in-app only.	FR-3.4, UC-8.0	Add push notifications.
Load time on mobile data not measured.	NFR-1	Measure on a phone over mobile data after deployment.
Prototype runs locally.	NFR-8	Deploy behind HTTPS.